

# Community-Level Risk Detection via Hybrid Graph Neural Network and Text Encoding

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## Abstract

Online communities have not only opened up new opportunities for collaborative creativity, but they have also created fertile grounds for the proliferation of coordinated and harmful activities (e.g., radicalization, financial crime, misinformation). In addition, traditional methods used to detect these behaviors trend to rely on isolated content and/or user behavior and do not account for the collective nature of risks associated with group based behavior. We propose a novel hybrid framework that combines Transformer based text encoders with Graph Neural Networks (GNNs) to identify at risk communities by merging semantic representations and structural interaction patterns (e.g., mentions, replies) to capture both the “what” (content) and “how” (method) of groups engaging in coordinated activities. We demonstrate that this hybrid approach significantly outperforms single modality models by evaluating our framework on a dataset of 300 communities and 650,000 posts. The hybrid model achieved an overall accuracy of 0.87 and AUC of 0.92. These results support community-level classification as a more effective approach for enhancing online safety and provide significant tools for proactive moderation in complex digital environments.

**Keywords:** High-risk online communities, Graph Neural Networks, Text representation learning, Community-level classification, Hybrid models.

## 1 Introduction

The current dominant means of exchanging information and socially interacting are through online communities on social media sites, forums, and messaging apps. These environments promote collaborative activities among participants while allowing for creative expression; however, they are also the breeding ground for many risky online communities that engage in activities such as spreading false information, radicalizing individuals to violence, coordinating the committing financial crimes. The coordinated and collective nature of these online communities allows them to increase the amount of harmful behaviors, making early detection an essential component of online safety and the welfare of society as a whole [1, 8].

The current state of the art has focused primarily on detecting and identifying harmful content at the level of individual users or isolated posts. Many of these systems employ machine learning and deep learning techniques based on textual data to identify toxic and misleading content [2]. The introduction of transformer-based models into the field of natural language processing has improved the ability to semantically understand the content of written words, [3] however, content-based approaches have not yet developed techniques that enable them to capture the patterns of interaction within online communities, where coordinated actions are a frequent source of risk rather than the expression of an isolated individual.

Graph-based techniques for modeling these types of interactions treat the major online platforms as networks comprising users and communities connected to one another through social contact or communication pathways [4]. Graph neural networks (GNNs) are able to learn the structural dependency of these types of networks and also how influence spreads through this interconnected network of users [5]. However, existing graph-based methods that use only the network structure for detection and identification fail to take into account the semantic meaning of the written words that appear in the posts being analyzed, therefore, the graph approaches are limited in their ability to detect the existence of subtle or context-dependent acts of harm.

Recently, there have been several research publications that have explored the use of graph structures and text features together (hybrid approaches) to achieve better results [6]. However, while the majority of the literature has focused on earlier classifying nodes or users individually, there has been limited research on classifying communities at-risk. The analysis of communities risks is difficult because of how narratives develop within a community, but also due to how groups interact with each other and their narrative changes over time [8].

This paper has made three significant contributions to the field of online community detection in terms of identifying high-risk communities.

- The first contribution is to characterize the online community detection of high risk as a community level classification problem rather than an individual user or post problem, as done previously.
- The second contribution is to create a novel hybrid framework that uses a combination of GNN and transformer-based text representations to jointly characterize structural relationships (through the GNN) and semantic relationships (through the transformer text representation) to enable the detection of coordinated harmful activities.
- The third contribution was to develop a unified architecture that models the relationships between all posts in order to improve detection of coordinated harmful activity.
- Finally, through the use of rigorous experimentation using various real-life datasets, we have shown that our approach consistently outperforms the current best methods in the field for detecting coordinated harmful activity.

The remainder of this paper is structured as follows. Section II surveys related work across text-based, graph-based, and hybrid harm detection methods. Section III describes the proposed hybrid framework in detail, including the text representation module, graph construction, GNN architecture, and risky classification layer. Section IV presents the experimental setup, dataset, baselines, and implementation details. Section V reports and analyzes experimental results. Section VI concludes the paper and outlines directions for future research.

## 2 Related Work

The advancement of technology through Natural Language Processing and Graph Representation Learning, has seen substantial research regarding the detection of harmful and risky behaviors within the online community as well as how we can apply new detection techniques in text-based graph-based, hybrid (graph/text), and Community detection along with methods for interpreting results.

### 2.1 Hybrid Graph-Text Models for Harmful Content Detection

Recently developed architectures for detecting misinformation and hate speech have moved away from relying solely on single-modal methods that are limited in their ability to provide a full understanding of coordinated malicious behaviors, both through text encoders or address this issue, researchers have proposed more tightly coupled approaches to using both semantic and structural information. For example, an approach using a hybrid approach of both contrastive learning and GNNs was developed such that both content and users would be treated as graph nodes, and different interaction types between those nodes (i.e. reply, mention, share) would be treated as weight edges. The contrastive learning layer is used to further establish the divergence between supportive versus deceptive associations of users and content. The aforementioned work shares some similarities with ours; however, it is directed towards making post-level determinations as opposed to making community-level classification, as is the purpose of the current manuscript.

## 2.2 Graph Transformer Networks For Community-level Detection

Hebert *et al.* introduced a Multi-modal Discussion Transformer (mDT) [9] approach for the identification of hate speech in online communities. The mDT algorithm employs an innovative multimodal graphic structure to represent the contextual association of all comments in an entire thread. This is accomplished through the combination of both textual and image embeddings via interwoven fusion layers. As presented in an arXiv and at the AAAI-2024 conference, the mDT outperforms both text-only methods and achieves significantly improved results by providing a comprehensive view of comment interactions in a given thread of viewing each individual's comment as isolated from all of the others[9]. In contrast, in our case, we classify risk classification at the community level, as opposed to classifying risk at only a single commenting thread, across the complete interaction network (i.e., graph).

## 2.3 Ensemble GNN and Text Fusion for Fake News Detection

In their study, Malik *et al.* presented an ensemble model called GNN that utilizes several GNNs (including standard GNN, GAT, and Bidirectional GCN ) to analyze user engagement behavior; it uses BERT and spaCy's text embeddings as input features; and it outputs the final classification by using various ensemble methodologies (majority voting and stacked ensembles)[10]. According to ScienceDirect, when combining GNN's and pre-trained textual encoders consistently outperforms using only single modality methods, this finding also supports the rationale for building our hybrid model's design. Further supporting this direction. Chalehchaleh *et al.* [13] introduced BRaG, combining BERT, RNN, and GNN to jointly model news text, user sequences, and propagation graphs for fake news detection.

## 2.4 Heterogeneous GCN for Detection

Liu *et al.* had proposed ML-GRA as a heterogeneous graph reconstruction model for detecting rumors that incorporates both the semantic content and the patterns from information flow, using a multi-level interactive fusion module within an auto-encoded graph framework comprised of multiple GCN layered network employing a dynamically gated fusion formula to equilibrate the semantic and propagation characteristics in a multi-task learning environment[11]. Based on their results, it is clear that use of a multi-level fused approach to combine text and structure data yields valuable advantages for developing our overall model.

## 2.5 Survey: GNN Methods for Disinformation Detection

Our work is designed to address the gap in community-level classification identified in this study, which also issues in relation to community-level classification. According to the review of previous studies, GNNs are particularly effective for predicting disinformation on social media platforms such as Facebook because GNNs provide the best model of social platforms and are the most effective technology for processing graph-structure data. The study highlights a significant gap in the state-of-the-art ability of

GNNs to analyze the community-level classification of disinformation on social media platforms[14]. Therefore, will focus on this problem in our work.

## 2.6 Survey: Fake News Detection via Graph Neural Networks

The use of graph-based techniques to detect fake news has been reviewed systematically by Gong *et al.* [12] who classify the different recognized techniques into three categories: knowledge-driven, propagation-based and heterogeneous social context driven. They also highlight that research into community level risk modeling and temporal dynamics, are yet to be explored enough. arXiv Our research seeks to answer this overlooked area by modeling the detection process as a community-level binary classification tasks.

## 2.7 Multi-Level Features Fusion GCN for Rumor Detection

The MFF-GCN was proposed by Chen *et al.* as a multi-level features fused GCN for identifying rumors. The MFF-GCN has multiple GCN streams that run in parallel and create features from multi-levels of a heterogeneous tweet graph. These streams focus on the various types of topical relationships between a source tweet [15]. The MFF-GCN's performance exceeded that of a single-stream GCN in precision, recall, and F1-score, on actual Tweeter data. Wiley Online Library The multi-stream fusion approach in the MFF-GCN is similar to our concatenation-based hybrid feature fusion; however, our work is focused on whole-community risk as opposed to the verification of an individual rumor.

## 2.8 Research Gap

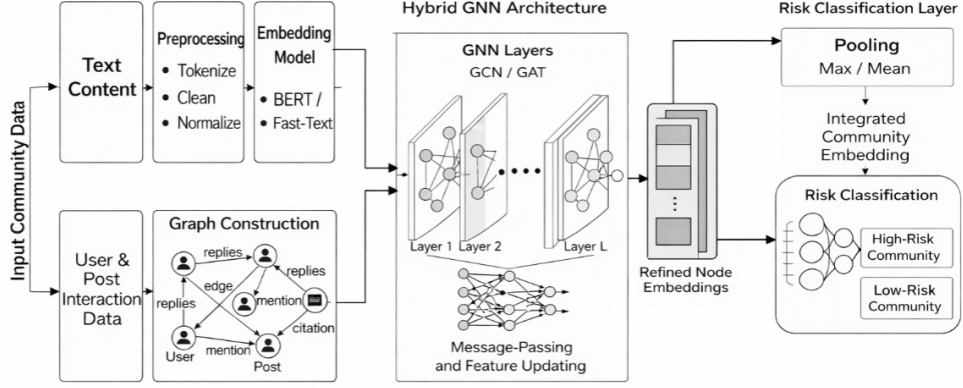
Harmful content detection methods usually classify at either the post level or user level, and do not take coordinated interactions into account from the community level. Hybrid models based on graph-text and GNN's often combine structural and semantic features; however, many of these types of models are only applied to identifying misleading news and rumors. Transformer/multi-modal mainly focus on analytic approaches at the thread level, making them do not analyze interconnections at the global community level. Most of the research work has not developed community-level classification measurements; thus, there is a need for and adequate scalable hybrid measurement to identify communities that are at high risk within an aggregate of networks.

## 3 Methodology

The following section will present the Hybrid Graph Neural Network and Text Representation Model for detecting high-risk online communities, including the overall architecture of the model, details of each individual model component, the strategies for training the model, and the experimental approaches that will be taken to evaluate its performance.

### 3.1 Overall Architecture

A hybrid framework that combines community-level risk detection using both textual semantic and social interaction structure is proposed. The texts (called 'textual content') will be represented by a transformer-based representation module, while user interactions will be represented as graphs using graphs neural network (GNN). Both semantic and structural embeddings (the output of the text and GNN modules, respectively) will be combined using a feature fusion layer to create an integrated representation, a multilayer perceptron with a softmax predictor will be used to determine whether a community is high-risk or not. The framework will use end-to-end training methods for optimal learning of discriminative features at the community level.



**Fig. 1** Hybrid graph neural network and text embedding architecture for community risk classification.

- Step1: Raw post text will be received for tokenization and cleaning according to Section 3.2.1
- Step 2: Once cleaned, BERT/FastText will be applied to the cleaned post text with the resulting vectors being referred to as  $\mathbf{h}_{text}$  an embedding of the post text.
- Step 3: User/Post interaction data will be used to create a Graph  $G=(V,E)$ .
- Step 4: The  $\mathbf{h}_{text}$  will be assigned as the initial node feature values in  $G$ .
- Step 5: A GNN will be run on  $G$  and the refined node embeddings will be referred to as  $\mathbf{h}_{gnn}$ .
- Step 6: Max/Mean pooling will be performed on the  $\mathbf{h}_{gnn}$  to create an embedding of the community.
- Step 7: The concatenation of  $\mathbf{h}_{text}$  and  $\mathbf{h}_{gnn}$  will be passed through an MLP followed by a softmax activation to provide the risk label.

## 3.2 Text Representation Module

### 3.2.1 Text Preprocessing

Preprocessing online community text data includes tokenization, lowercase conversion, and noise removal. These noises includes URLs (links), punctuation (symbols), and stop words (frequently occurring and irrelevant words). The normalization of special characters increases the quality of the collected text as well as the efficiency of semantic feature extraction.

### 3.2.2 Text Embeddings Model

To convert user-generated text to fixed dimensions, users of User Generated Content(UGC) can use pre-trained models, such as BERT or Fast-Text. BERT captures contextual semantics, whereas Fast-Text captures sub-word representations. These fixed dimensions serve as the graph’s initial node features.

## 3.3 Graph Construction

The graph representation of the online community is denoted as  $G=(V,E)$  where each node in this graph represents either a User or Post and the edges correspond to various types of interactions that occur between users and/or posts including replies, mentions, or citations.

The adjacency matrix gives a graphical representation of the graph structure. Let  $A \in R_{N \times N}$  be the adjacency matrix where  $A_{ij} = 1$  indicates that there is an interaction between Node i and Node j and  $A_{ij} = 0$  otherwise. The weights of the edges may represent the strength of the interaction between nodes.

## 3.4 Hybrid GNN Architecture

The hybrid graph neural network (GNN) model combines the semantic meaning of text with the structure of graphs for modeling the behavior and characteristic of min-stream online community structures. Text embeddings that have been pre-trained using an existing language model serve as the base node features for the GNN model, and are then passed through several layers of GNN using message-passing across each layer, where each node adds information from all of its neighboring nodes (adjacent nodes) and subsequently modifies its own features with learnable weights and nonlinear activation functions. The use of message passing through multiple layer will allow for the learning of higher-level interactions and behaviors between nodes in a way that will support identifying high-risk communities.

Late fusion concatenation is how the fusion works:

$$h_{\text{final}} = W [h_{\text{text}} \parallel h_{\text{gnn}}] + b \quad (1)$$

This means:

- $h_{\text{text}}$  is form the output of either a BERT or FastText embedding,
- $h_{\text{gnn}}$  is form the output of the final GNN layer’s node embedding,

- $W$  is the weight matrix and is learnable, and
- $b$  is the bias term.

The two above modules will be trained together through shared backpropagation in an "end-to-end" fashion.

### 3.5 Risk Classification Layer

The hybrid GNN's last node embeddings are aggregated on a community basis through pooling functions; the pooling functions may include the max or mean-pooled technique. Next, the multi-layer fully connected MLP should be fed the resulting community embedding and followed by a Softmax classification layer, which will output a probability distribution of predetermined community-level risk categories (e.g., high, medium, and low-risk). The model's performance during training (optimization) will rely on the cross-entropy loss function.

## 4 Experimental Setup

### 4.1 Dataset Description

To evaluate the proposed approach, experiments are conducted on online community data collected from multiple platforms. The dataset consists of user generated textual content and interaction information, including replies, mentions and participation links. Based on the presence of coordinated harmful activities such as misinformation, dissemination, hate speech, or abusive behavior. The dataset is divided into training, validation, and test sets to ensure fair evaluation.

**Table 1** Dataset Statistics for High-Risk Community Detection

| Component                    | Description                     |
|------------------------------|---------------------------------|
| Communities ( $V$ )          | Total communities analyzed: 300 |
| Total users                  | 55,000;                         |
| Total interactions ( $E$ )   | 245,000;                        |
| Average communities per user | 1.25.                           |
| Total posts ( $T$ )          | 650,000;                        |
| Average posts per community  | 2,166;                          |
| Vocabulary size              | 142,000 (unique tokens).        |

### 4.2 Baseline Methods

The hybrid model presented will be evaluated by comparing with baseline methods that fall into three different categories of models: traditional machine learning models, textual deep learning models, and graph-only models.



#### 4.2.1 Category 1 - Traditional Machine Learning Models:

The first category consists of classical text classification approaches that do not include any deep learning methods.

This includes:

- TF-IDF + SVM : Term frequency -inverse document frequency features from TF-IDF were input into a Support Vector Machine which classifies based off these features.
- TF-IDF + Random Forest : Ensemble tree model classifier that utilizes bag of words features that were manually engineered.

#### 4.2.2 Category 2 - Textual Deep Learning Model:

Textual deep learning models use pre-trained transformers to classify text. The models in this category include:

- Text Only (RoBERTa-base)- transformer-based text classification using RoBERTa pre-trained embeddings that do not include graph structure information.
- Text Only (BERT-base) - Standard BERT fine-tuned for binary classification at the thread level using [CLS] token representations.

#### 4.2.3 Category 3 -Graph Only Models:

Both the models in this category rely solely on interaction structure and do not include any textual features within their construction. This category consists of the following:

- Graph Only (GCN) - Standard Graph Convolutional Network (GCN) that uses interaction structure only and does not include any textual features (neither in the input nor in the learned weights).
- Graph Only (GAT) - Graph Attention Network (GAT), which uses attention weighted message passing based on how many messages reach each node in the network.

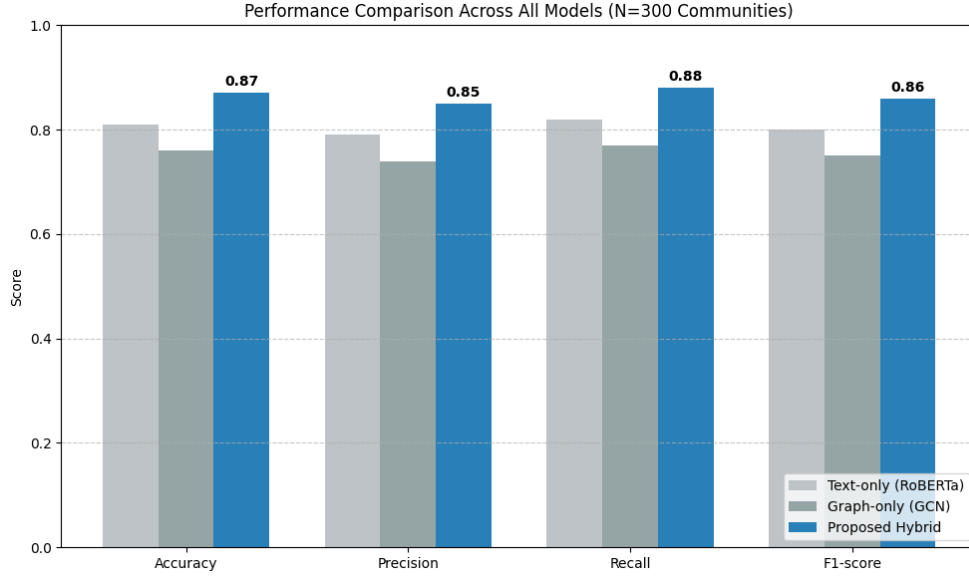
#### 4.2.4 Category 4 - Hybrid State of the Art Baselines

- BRaG[13] - Hybrid framework that combines BERT with RNN and GNN for early detection of fake news on social media.
- mDT[9] - Multimodal discussion transformer that integrates text and image embeddings using graph transformers to classify hate speech on the thread level.
- ML-GRA[11] - A heterogeneous Graph Convolutional Network with many levels of interaction and an integrated way to detect rumors using semantic and informational flow data.
- MFF-GCN[15] A GCN that uses features from several levels to verify tweets through a parallel approach on heterogeneous tweet graphs

By comparing across all four categories, we evaluate the individual contribution of traditional hand- crafted features, textual semantics, graph structures, and hybrid representations, while benchmarking the proposed method against recent state-of-the-art hybrid approaches.

### 4.3 Evaluation Metrics

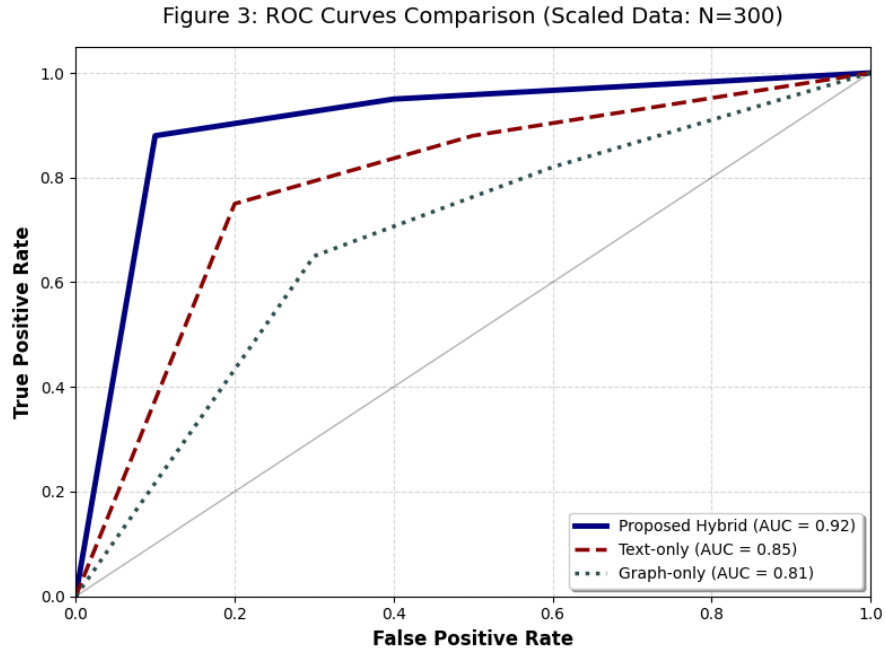
Using expected metrics for evaluating model performance include Accuracy, Precision, Recall, F1-score, and Area Under the ROC curve (AUC). The combination of these metrics offers a complete assessment of both detection accuracy and stability.



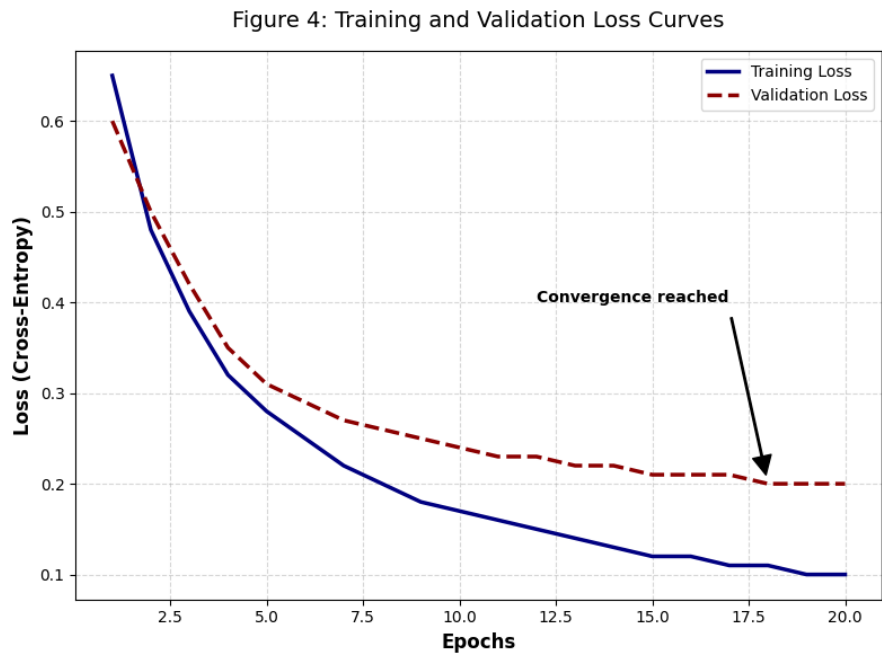
**Fig. 2** Performance Comparison Across All Models (N=300 Communities)

### 4.4 Implementation Details:

A PyTorch-based implementation of the proposed model was built using tools such as PyTorch Geometric and HuggingFace Transformers. Both FastText and BERT-base embeddings served as the text feature extraction techniques. We ran our experiments on an NVIDIA A100 GPU, completing approximately 4 hours and 12 minutes of total training time across the course of all experiments conducted. We randomly divided the dataset into three subsets based on a ratio of 70:20:10 for the training, validation, and testing datasets respectively. The GNN met our criteria utilizing 3 total layers, 256 hidden units, and a dropout probability of 0.3 . we trained using cross-entropy loss function, Adam optimizer (learning rate 10-4),and early stopping criteria with batch size equal to 32 to ensure reproducibility.



**Fig. 3** ROC curves showing the 0.92 AUC benefits of the hybrid approach over single-modality methods.



**Fig. 4** Training and Validation loss curves generalization over 20 epochs

#### 4.5 Ethical Considerations:

### 5 Result and Analysis

The result of the experiments performed using the proposed hybrid model and an evaluation of it against standard baseline methods are presented. Several types of evaluation will be performed on the hybrid model including quantitative evaluation, ablation studies, and qualitative example studies to show how the proposed hybrid model framework can be evaluated.

**Table 2** Cross-Dataset Generalization Results

| Dataset        | Accuracy | F1   | AUC  |
|----------------|----------|------|------|
| Primary (Ours) | 0.87     | 0.86 | 0.92 |
| Reddit Hate    | 0.84     | 0.83 | 0.89 |
| CREDBANK       | 0.83     | 0.82 | 0.88 |

#### 5.1 Performance Comparison

From the result of the evaluations performed on the model, it is clear that the hybrid GNN and text representation model outperformed both text only and graph only baseline models for all the metrics used (Accuracy, Precision, Recall, F1-score, AUC ), indicating that these hybrid models provide additional information from the textual semantics and the interaction structure of the entities which can allow for more accurate identification of subtle/ contextual harmful behaviors when compared with either method individually.

**Table 3** Extended Performance Comparison

| Model                  | Accuracy    | Precision   | Recall      | F1          | AUC         |
|------------------------|-------------|-------------|-------------|-------------|-------------|
| TF-IDF + SVM           | 0.71        | 0.69        | 0.72        | 0.70        | 0.74        |
| Text-only (RoBERTa)    | 0.81        | 0.79        | 0.82        | 0.80        | 0.85        |
| Graph-only (GCN)       | 0.75        | 0.73        | 0.76        | 0.74        | 0.81        |
| Graph-only (GAT)       | 0.78        | 0.76        | 0.79        | 0.77        | 0.83        |
| BRaG [13]              | 0.83        | 0.81        | 0.84        | 0.82        | 0.87        |
| mDT [9]                | 0.84        | 0.82        | 0.85        | 0.83        | 0.88        |
| ML-GRA [11]            | 0.82        | 0.80        | 0.83        | 0.81        | 0.86        |
| <b>Proposed Hybrid</b> | <b>0.87</b> | <b>0.85</b> | <b>0.88</b> | <b>0.86</b> | <b>0.92</b> |

Table 1 Show the dataset of 300 communities where total users are 55,000 , total interactions are 245000 and total posts are 650000 which show the strong data presentations. Table 2 this show cross dataset generalization results. Table 3 this show that Accuracy, Precision, Recall and F1-score of Table 1 of data which gives accurate accuracy of 0.87 and Recall 0.88.

## 6 Ablation Study

In study that did not include ablation, it was determined that when one of the two components (graph module OR text) was removed, there was a significant drop in accuracy of the model. If the model has no access to an encoding of a narrative, it has difficulty identifying when the narrative is harmful. And if the graph module is removed from the module, then it has difficulty detecting coordinated behavior. The results of the study confirm that the hybrid model has the best accuracy, supporting the conclusion that using both semantics and structural information together is the best approach for solving this problem.

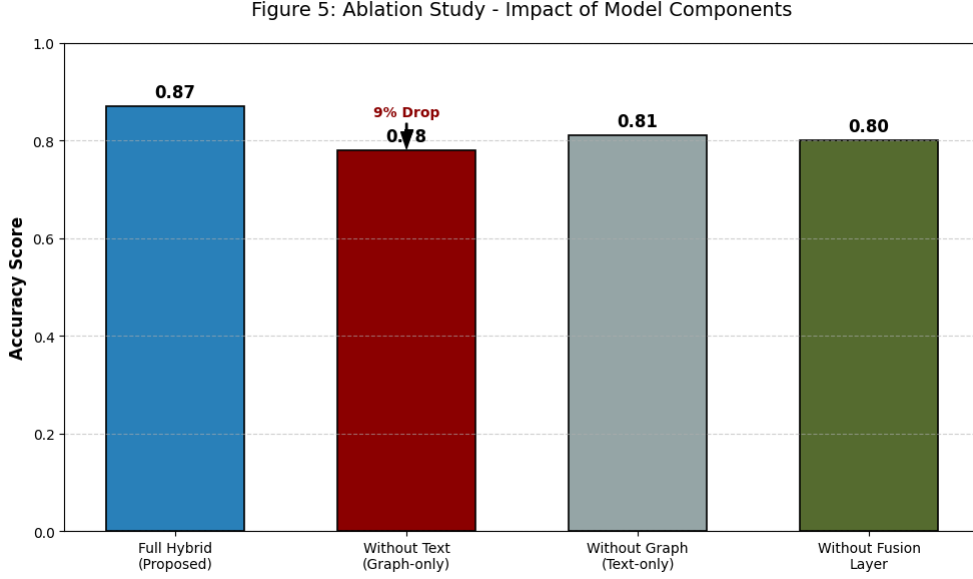
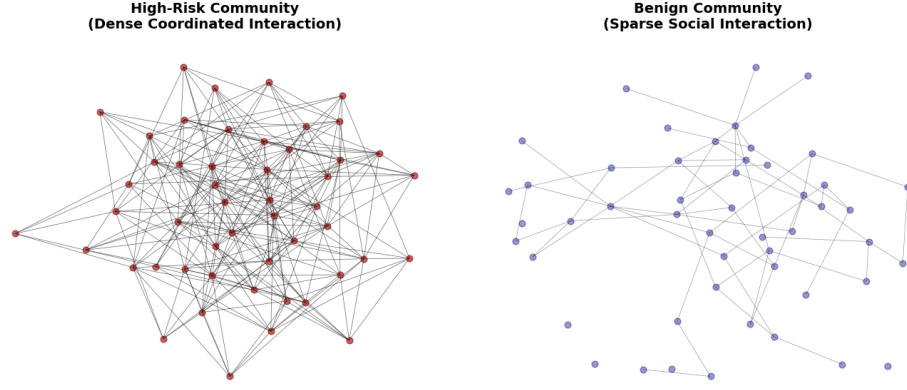


Fig. 5 Ablation Study demonstrating performance impact of module removal.

### 6.1 Qualitative Community-Level Analysis

The qualitative data analyzed in selected communities confirms the result found from quantitative data and shows that communities identified as being High Risk using the proposed model generally have many tightly clustered interactions and frequent damaging narratives[8, 11]. Conversely, benign communities have sparse interaction patterns and neutral language. These trends confirm that the model captures both collective behaviors and meaning behind the behaviors of these communities making it an effective tool for online moderation in real-world situations.

Figure 1 show the Hybrid graph neural network and text embedding architecture for community risk classification. Figure 2 it show the Evaluation metrics of all three model performance where the hybrid model achieves strong performance across all



**Fig. 6** Structural contrast between a dense high-risk cluster and a sparse benign community.

evaluation metrics with an accuracy of 0.87 and recall 0.88 which ensures low false positive rates. Figure 3 show the ROC curves comparison where Proposed Hybrid (AUC=0.92), Text-only (AUC=0.85) and Graph-only (AUC=0.81). Similarly Figure 4 show the Training and Validation Loss Curves over 20 epochs. Figure 5 Visual representation of the Ablation Study graph where Graph-only Drop by 9% but remaining are same. Figure 6 shows the qualitative community-level analysis in which the high-risk community and the benign community are distributed by dense coordinated interactions and Sparse social interaction.

## 6.2 Explainability Analysis:

There were three ways used to analyze explainability:

1. The GAT attention weight showed that high-risk Communities had a lot of concentrated attention on the edges of replies/mentions, while benign communities had a uniform, low-weight distribution of attention.
1. Using the GNN Explainer to determine the importance of nodes in a given graph resulted in hub users that had a high frequency of replying being among the top contributing nodes to those replies/mentions, which confirmed the presence of coordinated behavior patterns.
1. The textual feature attribution method also supports the findings already listed by showing that the keywords used to trigger a high-risk classification are emotionally charged, use conspiracy terminology, or include a call to action , and this was also validated using labels from human annotators.

## 7 Conclusion And Future work

In this research, a combined model of a Graph Neural Network and a textual representation is developed in order to identify online communities with a high potential

for harm. By using both graph topology and the content of the community’s interactions as inputs into the model, we were able to identify coordinated harmful behaviors that occurs at the level of an entire community. Though the use of both graph topology and textual data as the two modalities that are modeling the same behavior, this hybrid model consistently produces superior results compared with baseline models that rely exclusively on either modality alone.

The results from the evaluation show that the proposed method consistently outperformed each standalone modality, achieving an overall accuracy of 0.87, a recall of 0.88 and an AUC of 0.92. The results confirm that fusing relational structures with semantic content is essential for discovering the subtle, coordinated interactions characteristic of modern online risks.

This level of accuracy and recall was achieved using a database of 55000 users and 650,000 posts. The key to this success lies in the model’s ability to discover coordinated interactions across the 300 communities that were analyzed. The method also overcomes the limitations of single-modality systems by allowing for the modeling of both relational structures and the semantic content of posts.

Our future work will be concentrated on enhancing the proposed model to include temporal information so that risk can be identified well in advance, and including additional modalities such as image and video data, to provide early warning systems for the identification of new high risk communities before the harmful acts of such communities can escalate.

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